STEAMROLLER

Inclusive ASR for People Who Stutter. A multi-agent system designed to transform stuttered speech into fluent output in real-time.

Presented by - Aditya Suryakar

Satyjeet Patil

Sanika Sathe

Pranali Sawant

Prachi Yargatti

Under the guidance of : Prof.Mrs.R.S.Kamble



Problem Statement

Stuttering(difficulty in speaking smoothly) is a multifaceted fluency disorder that disrupts speech flow and hampers communication. Its origins remain unclear, involving both biological and psychological factors , often leading to social isolation and emotional distress.

Project Objective

This project aims to design an inclusive, real-time AI system that:

- Converts stuttered speech into fluent speech.
- Preserves the original semantic meaning.
- Produces natural, stutter-free audio.
- Works efficiently with low latency.

Challenges

C#1: Real-time Speech Transformation : Real-time transformation of stuttered audio into fluent output is complex, especially as current speech modeling technologies are primarily trained on fluent speech.

C#2: Preserving Semantic Integrity : It is crucial to prevent semantic loss during transcription, ensuring the system accurately preserves the speaker's intended message despite stuttered speech.

C#3: Ensuring Low Latency : For natural, uninterrupted communication, the speech repair process must operate with extreme efficiency and minimal latency.

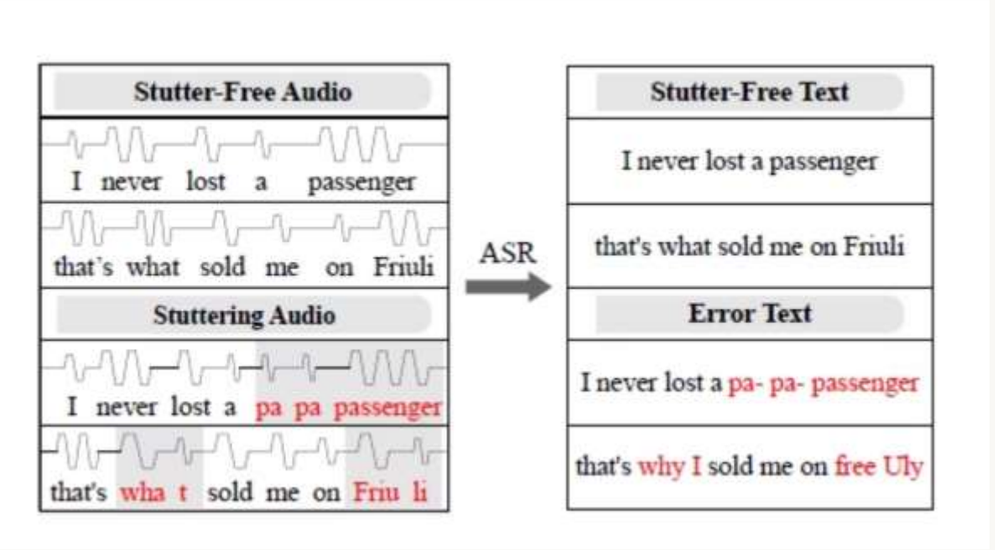


Figure 01 : Comparison of ASR transcription outcomes for stutter-free and stuttering audio inputs.

Existing Solutions

Conventional ASR Systems

Conventional Automatic Speech Recognition (ASR) systems such as Google Speech-to-Text, Microsoft Azure Speech, and Whisper are trained mainly on fluent speech. These systems struggle to accurately recognize stuttered speech, leading to high error rates and loss of meaning.

Speech Therapy & Clinical Treatments

Speech therapy and clinical treatments aim to reduce stuttering over time, but they require long-term effort and do not provide immediate support during real-time communication like meetings or interviews.

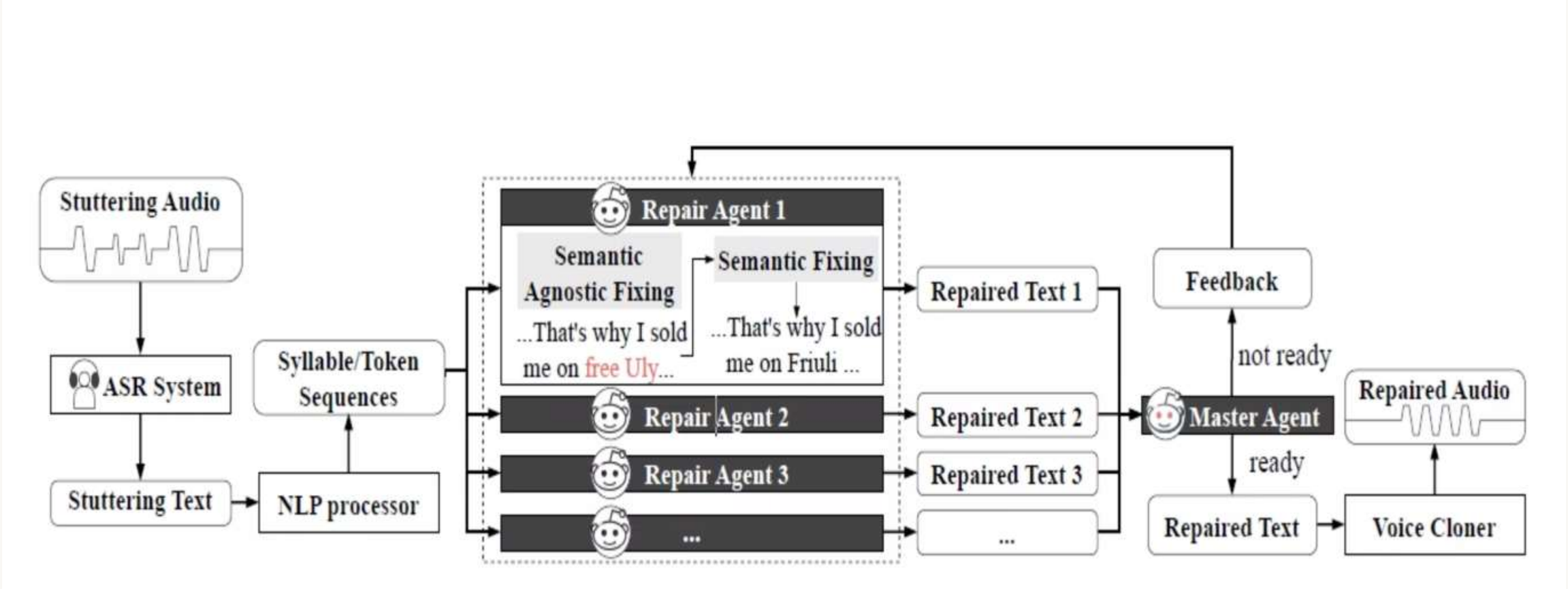
Stuttering Detection & Classification

Stuttering detection and classification systems focus on identifying disfluencies such as repetitions and blocks, but they do not correct or repair the speech content.

Electronic Fluency Devices

Electronic fluency devices alter auditory feedback to improve fluency, but they are expensive, hardware-dependent, and not suitable for scalable software solutions.

WorkFlow


Stuttering Audio
ASR System
Stuttering Text
NLP processor
Syllable/Token Sequences
Repair Agent 1
Semantic Agnostic Fixing
...That's why I sold me on free Uly...
Semantic Fixing
...That's why I sold me on Friuli ...
Repair Agent 2
Repair Agent 3
...
Repaired Text 1
Repaired Text 2
Repaired Text 3
...
Feedback
not ready
Master Agent
ready
Repaired Text
Repaired Audio
Voice Cloner

Future scope

- **Integration with Online Meeting Platforms :** The system can be integrated with platforms like Zoom, Google Meet, and Microsoft Teams to provide real-time fluent speech output for people who stutter during virtual meetings and interviews.
- **Live Caption and Speech Repair Support :** STEAMROLLER can be used to generate real-time corrected captions and stutter-free audio in online meetings, improving understanding for all participants.
- **Improved Voice Naturalness :** Future improvements in voice speech thcloning can help generate at more closely matches the user’s original voice and emotions.
- **Multi-Language and Accent Support :** The system can be extended to support multiple languages and regional accents, increasing its global usability.
- **Real-Time Performance Optimization :** Further optimization can reduce latency, making the system suitable for live conversations and instant responses.
- **Mobile and Edge Deployment :** The solution can be deployed on mobile devices or edge systems for better accessibility and privacy.

Research Papers

- STEAMROLLER: A Multi-Agent System for Inclusive Automatic Speech Recognition for People who Stutter
- Automated Dynamic Stuttered Speech Recognition System and Conversion System using Mel Filter And Enhanced Logistic Regression Model
- Speech Disfluency Removal and Augmentation for Enhanced Voice Technology Interaction

Conclusion

STEAMROLLER provides a novel framework for converting stuttering speech into fluent audio, offering dual benefits:

- Seamless audio for PWS.
- Enhanced tools for developers to improve stuttered speech recognition and annotation.

This work paves the way for more inclusive AI systems, ensuring that voice technology serves everyone.